

Collapse-Aware AI (CAAI):
A Computational Framework for Memory-Conditioned Drift Regulation in Neural Inference Systems

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ABSTRACT

This paper presents Collapse-Aware AI (CAAI), a computational framework for adaptive bias regulation in neural inference systems.

The model recasts earlier formulations of Verrell's Law into a stochastic drift–diffusion process, defining how informational bias evolves dynamically within logit-space during inference.

Unlike static entropy controls or RLHF, CAAI incorporates a memory-conditioned prior directly into the drift term b_ψ , with a corresponding Governor module that regulates drift and diffusion in real time.

1. POSITIONING

Modern neural language models generate output by sampling tokens from probability distributions shaped by large-scale training. These systems lack persistent memory, dynamic bias regulation, collapse stability checks, and continuity. Collapse-Aware AI treats inference as a controlled collapse rather than a raw token-sampling event. It introduces a memory-conditioned prior, a dynamic drift term b_ψ , and a real-time Governor to ensure stability.

2. INTRODUCTION

Large models perform statistical continuation but do not regulate collapse or track uncertainty. This leads to oscillation, drift, contradictory reasoning, and high-entropy failures. CAAI reframes inference as:

$$dz_t = b_\psi(z_t, M_t) dt + S dW_t$$

with a composite drift term:

$$b_\psi = \alpha \nabla_z \log \pi_{\text{prior}}(z|M_t) + \beta \nabla_z \log \pi_{\text{anchor}}(z) - \gamma \nabla_z H(p_t)$$

The drift is recalculated dynamically, making the system adaptable and stable over long sessions.